PROJECT DOCUMENT

Mitsue Project

Research Brief

Version	v1.2
Date	2026-06-21
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MITSUE.IT PROJECT

Research Brief: Mitsue Village

Emerging Industries, Rural Revitalization & the Data-Center Opportunity

Compiled May 2026 | For internal project use

> **"The forest our ancestors planted — powering the village they built."** 先人が植えた森が、今、村に豊かさをもたらす。

1. Village Profile & Demographics

Mitsue-mura (御杖村) is a mountain village in Uda District, Nara Prefecture, located on the southern Soni Plateau along the upper Nabari River. The village sits on the border with Mie Prefecture and is part of the Muro-Akame-Aoyama Quasi-National Park.

Indicator	Figure	Unit	Year
Population	1,393	persons in 782 households	Apr 2024
Population density	18	persons / km²	Apr 2024
Total area	79.58	km²	—
Forest cover	88–90%	% of total area	current
Avg annual temperature	12.5	°C	—
Avg annual rainfall	2,174	mm	—

The village has no railway service. The nearest station is Nabari (Kintetsu), approximately two hours from Osaka by rail and bus. The official village catchphrase is “Rich in historical romance and nature’s bounty.”

Source: Wikipedia / Mitsue Village official website

2. Current Economic Base

Agriculture and forestry dominate Mitsue’s economy. While traditional in character, the village faces structural decline that mirrors prefectural and national trends in rural Japan.

Forestry

Nara Prefecture’s overall forestry output declined from a peak of ¥351 billion in 1975 to just ¥26 billion in 2022 — a drop of roughly 93% in real terms. Mitsue’s forests, covering 88% of the village area, represent the village’s largest undeveloped asset.

- 2022: Regional Revitalization Cooperation Corps program managed 27.81 hectares of forest annually, funded by the forest environment transfer tax.
- 2022: Two new forestry entrants recruited. Five-year target: four additional workers, 100 hectares under active forest management.
- Goal: establish self-felling forestry models suitable for small-scale operators.
- International export initiative: model wooden homes being constructed in Thailand using traditional Japanese joinery techniques, targeting markets where wooden architecture has largely disappeared.

Source: Grokipedia / Mitsue Village planning documents

Agriculture

Agricultural output was ¥5.0 billion in 2022, led by vegetables (¥2.4 billion) and beef cattle (¥1.3 billion). The village supports new entrants through facility vegetable production subsidies and drone-assisted farming programs, aligned with “sixth industrialization” principles (linking primary production with processing and retail).

- Free childcare and fully subsidized school lunches for families.
- Greenhouse and agricultural machinery purchase subsidies for new farmers.
- Newcomers profiled: spinach and komatsuna growers Datedo Shohei and Kimura Yukihiro joined via the Cooperation Corps.

Source: MynaviAgri / Grokipedia

Manufacturing & Services

Small-scale manufacturing is minimal. Shipments reached ¥538 million in 2018 — down approximately 94% from ¥8.7 billion in 1975. Tourism assets include the Himeishi-no-Yu hot spring roadside station, Mount Miune (one of Japan’s 300 famous peaks), and the historic Ise Honkaido pilgrimage route.

3. Migration & Settlement Programs

The village actively recruits new residents and workers through a layered support system combining national programs with village-specific incentives.

Trial Stay Housing

Fully furnished and appliance-equipped move-in-ready housing is available to relocation candidates at ¥2,000/night for stays of 7–30 days. Located walking distance from Himeishi-no-Yu roadside station with hot spring, farm shop, and basic services.

Multi-Generational Co-Residence Subsidy

Up to half of construction or renovation costs covered, capped at ¥1 million for families with children under school age (or applicants under 45), and ¥500,000 otherwise. Targets three-generation and two-generation households.

Regional Revitalization Cooperation Corps (地域おこし協力隊)

Nationally, 7,910 members were deployed in FY2024 across agriculture, forestry, digital transformation, education, tourism, and environmental conservation. Approximately 70% of corps members settle permanently in their host community after their tenure. Mitsue has been participating since around 2016.

Empty House Bank

A vacant property registry connects prospective residents with available housing stock throughout the village.

Source: Tabisumu / Mynavi / Ministry of Internal Affairs (MIC)

4. The National Data Center Opportunity

This section provides strategic context directly relevant to the Mitsue AI data center and biomass energy proposal.

Market Scale

Japan is the second-largest data center market among developed countries after the United States. Key projections:

- Market size reached USD 23.4 billion in 2024; projected to grow at 6.7% CAGR to USD 33.4 billion by 2030.
- Electricity consumption projected to more than triple: from 19 TWh in 2024 to 57–66 TWh by 2034.
- Peak demand expected to reach 6.6–7.7 GW by 2034 — roughly 4% of Japan's total national peak load.
- Generative AI market in Japan projected to grow 15-fold by 2030; AI infrastructure demand to nearly triple.

Source: JLL Research 2025 / Wood Mackenzie / IMARC Group

The Geographic Mismatch — Why Rural Sites Are Now Policy

This is the single most important structural argument for the Mitsue proposal:

- 90% of Japan's existing data centers are concentrated in Greater Tokyo and Greater Osaka.
- Most large-scale renewable energy sources — solar, wind, hydro, biomass — are located in rural regions: Hokkaido, Kyushu, and mountainous Honshu.
- AI training data centers have "relaxed low-latency requirements" compared to conventional ones, making rural siting technically viable.
- The Japanese government is actively promoting regional redistribution through "Regional Revitalization" policy and the "Digital Infrastructure Development Plan 2030."
- Local community opposition to urban mega-data centers is rising — protests have occurred in Hino (western Tokyo), Shiroi, and Inzai (Chiba).

Source: JLL Research / Mitsubishi Research Institute / Japan Times

Precedent: ZED ISHIKARI (Hokkaido, October 2024)

Japan's first 100% real-time renewable energy data center opened in Ishikari, Hokkaido in October 2024. Key design principles directly applicable to Mitsue:

- Powered by wind, solar, biomass, and natural cold-air cooling — no grid offsets or carbon accounting tricks.
- Cold outdoor air used for natural ventilation for more than half the year, eliminating conventional cooling systems.
- The model is described by operators as replicable in any region with similar climate or renewable resource characteristics.

Source: NoticiasAmbientales / KIOXIA Green Data Center initiative

Smart City & Society 5.0 Framework

The Smart City Institute Japan (SCI-Japan), established 2019, operates as a public-private-academic knowledge platform under the government's "Regional Revitalization 2.0" initiative. The program explicitly targets rural quality-of-life through digital technology across mobility, health, tourism, energy, environment, and governance.

Source: SCI-Japan / Smart City Institute Japan

5. Biomass Energy & Community Cohesion Research

Academic research on biomass-powered facilities in rural Japan is directly relevant to framing the Mitsue proposal beyond economics:

A peer-reviewed study (International Journal of Energy and Environmental Engineering, 2019) on biomass-heated community bathing facilities in rural municipalities found that the most significant benefit of local renewable energy deployment is not energy security or job creation per se, but the enhancement of residents' sense of attachment to their community. The authors explicitly criticize current municipal approaches for failing to capture this social cohesion dimension, arguing that community participation in renewable energy revitalization leads to "sustainable prosperity of the region."

A parallel academic study of sixth industrialization in Totsukawa Village (Yoshino District, Nara Prefecture — directly comparable to Mitsue) examines how forestry-wood industry integration and cooperative asset inheritance have built resilient rural economies. Totsukawa is a near-neighbor and valid model for Mitsue's own forestry ambitions.

Source: IJEE 2019 / ResearchGate / Springer Link

6. Broader Rural Revitalization Context

Mitsue's challenges are not unique — but that is a strength for proposal-writing, as it places the village within a nationally recognized crisis with established policy infrastructure.

- Nara Prefecture's population peaked at ~1.45 million in 1999 and has been declining since. Current population: ~1.3 million (2022).
- Rural areas of Japan typically have elderly populations of 30–40% of total.
- An estimated 900 towns and villages are projected to become unviable by 2040 at current depopulation rates (WEF / economists).
- Successive Japanese governments have failed to halt rural depopulation. The current Takaichi administration launched a new rural revitalization initiative in late 2025, with analysts saying success requires "a sweeping effort" beyond job creation — full quality-of-life transformation.
- Yoshino Cedar House (Nara Prefecture) is a cited example of successful Airbnb-supported rural revitalization run as a resident cooperative — geographically and culturally adjacent to Mitsue.

Source: WEF / Japan Times / Frontiers in Sustainable Tourism

7. Strategic Summary for Mitsue.it

The research above supports the following positioning for the project proposal:

Unique Convergence

Mitsue sits at the intersection of four nationally recognized priorities: rural depopulation response, native forest restoration and forestry revitalization, rural renewable energy resilience, and regional data center distribution. No other village in Uda District is visibly pursuing all four simultaneously.

Policy Tailwind

The Japanese government is actively subsidizing exactly the components of the Mitsue.it proposal: Regional Cooperation Corps recruitment, native forest restoration and forestry revitalization via forest environment transfer tax, digital infrastructure in rural areas, and decarbonized data center construction. The proposal is not ahead of policy — it is aligned with it.

Mitsue's own official Renewable Energy Plan. The strongest evidence of policy alignment is local: in January 2025 the village published its **"Plan for Maximum Introduction of Renewable Energy"** (御杖村再エネ導入最大化計画), funded by a Ministry of the Environment grant, with citable official figures — 9 kt-CO_2 village emissions (2021), **transport = 46%** (the largest sector), **~90% forest cover**, +2.1°C local warming over 42 years, a **60% reduction target by 2030**, and **carbon neutrality by 2045** (five years ahead of national). The plan explicitly calls for EV charging at public facilities, small-hydro, rooftop/carport and perovskite solar, V2H, VPP, distributed disaster-resilience, and a forest J-Credit system — and sets a target of **one resilient renewable+storage+EV site** in the village (currently zero). Producing the plan completed step 1 of the MoE decarbonization funding ladder, making the village eligible for the **地域脱炭素移行・再エネ推進交付金** (step 2): a 2/3–3/4 subsidy on solar/battery/EV/private-wire capex, paid to the municipality via public-private partnership. The Mitsue Project is positioned as the operating vehicle for this official plan. (See `mitsue_village_re_plan_alignment.md`.)

Sources: 御杖村再エネ導入最大化計画（概要版）, Mitsue Village, Jan 2025; MoE 地域脱炭素移行・再エネ推進交付金 https://policies.env.go.jp/policy/roadmap/grants/ · 実施要領 https://www.env.go.jp/content/900470616.pdf

The Key Argument

90% of Japan's data centers are in two metropolitan regions. 90% of Japan's forests are not. Mitsue has 88% forest cover, the available former Sugano Elementary School (Mitsue Taiken Koryukan; leading candidate site, final choice in Phase 1), existing biomass infrastructure (Mitsue Onsen boiler), and proximity to the Kintetsu Nabari corridor. The convergence is not coincidental — it is a design opportunity.

Comparable Precedents

- ZED ISHIKARI (Hokkaido): 100% renewable rural data center, October 2024.
- Totsukawa Village (Nara): sixth industrialization of forestry as regional anchor.
- Yoshino Cedar House (Nara): community-run tourism cooperative, UNESCO World Heritage area.
- Kamikatsu Zero-Waste Center (Tokushima): 80%+ waste recycling as rural identity and brand.

Prepared for Mitsue.it | Research compiled May 2026 | Sources listed per section